

Math 8 End of Q2 Test — Answer Key

Mary, Star of the Sea School · Dr. Park · Answer Key (Teacher Use)

1. Solve: $y = 3x$, $y = x + 6$. Find x . **Answer:** 3
2. Solve: $2x + y = 10$, $x - y = 2$. Find x . **Answer:** 4
3. Parallel lines system has how many solutions? **Answer:** 0
4. Coincident lines system has how many solutions? **Answer:** infinite
5. Rearrange $A = \frac{1}{2}bh$ for b . **Answer:** $b = 2A/h$
6. Rearrange $y = mx + b$ for b . **Answer:** $b = y - mx$
7. Rearrange $KE = \frac{1}{2}mv^2$ for v (positive). **Answer:** $v = \sqrt{(2KE/m)}$
8. Weighted avg: $70(0.3) + 90(0.7)$. Result? **Answer:** 84
9. Grades 0.25, 0.25, 0.5 with 80, 90, 100. Weighted avg? **Answer:** 92.5
10. Isotope avg: $80\% \times 24 + 20\% \times 25$. **Answer:** 24.2
11. $d = vt$: $v = 8$, $t = 5$. d ? **Answer:** 40
12. Free fall: $t = 2$ s, $g = 10$ m/s². d ? **Answer:** 20 m
13. Orbital period for Earth in hours? **Answer:** 24 hours
14. Substitute: $y = 3x + 1$, $y = 2x + 5$. Find x . **Answer:** 4
15. Eliminate: $3x + 2y = 12$, $x - 2y = 4$. Find x . **Answer:** 4
16. Rearrange $F = ma$ for m . **Answer:** $m = F/a$
17. 50% test at 90 + 50% HW at 80. Weighted total? **Answer:** 85
18. $v = 30$ m/s, $t = 4$ s. Distance? **Answer:** 120 m
19. Free fall: $t = 5$ s, $g = 10$ m/s². Distance? **Answer:** 125 m
20. System with infinite solutions: slopes and intercepts must be ____? **Answer:** equal (both)